VIOLA ETC

Exposure Time Calculator

Software Architecture and Physics Reference

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1 Introduction

VIOLA ETC (Vipa-based Instrument for Oxygen Loaded Atmosphere Exposure Time Calculator) is a Python-based tool that predicts the signal-to-noise ratio (SNR) achievable by the VIOLA spectrograph for a given target under specified observing conditions. It is accessible via a Streamlit web interface. The instrument is a near-infrared, ultra-high-resolution ($R = 300{,}000$) spectrograph, with the prototype covering the J (1.10–1.30 µm) and H (1.50–1.70 µm) bands; K-band (1.90–2.50 µm) coverage is under consideration. The ETC supports calculations for all three bands.

The ETC converts the target's apparent magnitude into a top-of-atmosphere flux density, propagates the flux through Earth's atmosphere modeled with the ESO SkyCalc tool (Noll et al., 2012; Jones et al., 2013), accounts for the relevant telescope and instrument characteristics, and computes the SNR per spectral resolution element. An optional module estimates the detectability of molecular species (CH_4, CO, CO_2, H_2O, NH_3, O_2, and OH) in exoplanet atmospheres using cross-correlation techniques (Snellen et al., 2015; Birkby, 2018; Snellen, 2025). The molecular templates used for this detectability estimate are generated with the NASA Planetary Spectrum Generator (PSG; Villanueva et al., 2018) for temperate planet classes and petitRADTRANS (Mollière et al., 2019) for hot planet classes. The relevant template subset is selected automatically based on the user's chosen planet class. Support for user-supplied sky models and molecular templates is planned for a future release.

1.1 Purpose of This Document

This document serves two purposes: (1) it describes the software architecture so that future developers can navigate, maintain, and extend the codebase; and (2) it provides a complete physics reference, covering the signal and noise equations, stellar SED and sky-background models, and the molecular template grid, so that the scientific basis of each calculation can be verified and revised. All equations reflect the implementation as it stands in v0.5.

1.2 Scope

The document covers VIOLA ETC version 0.5. The current implementation assumes a stellar point source: the slit throughput model and magnitude system are both derived under the point-source approximation. Extended sources (galaxies, nebulae) are not yet supported. This limitation will be addressed in a future release.

2 Software Architecture

2.1 Technology Stack

The application is built on Python 3.12 with Streamlit as the web framework. The full dependency list is pinned in `requirements.txt`.

2.2 Directory Structure

The project is organized as shown in Table 1.

Table 1: Project directory structure

`Path`	**Description**
`app.py`	Streamlit frontend: UI layout, session state, plotting, and run orchestration
`viola_etc/`	Core Python package containing all computation modules
`viola_etc/constants.py`	Physical constants, unit conversions, photometric zero-points
`viola_etc/models.py`	Dataclasses for inputs (`TargetParams`, `ObservingCondition`, `UserInputs`), hidden defaults (`InstrumentConfig`, `SiteConfig`), and the output container (`ETCResult`)
`viola_etc/core/runner.py`	Main ETC orchestrator: `run_etc()` stitches together the full pipeline
`viola_etc/core/validate.py`	Input validation functions for all dataclasses and sweep parameters
`viola_etc/observing/sky_io.py`	Sky model I/O: FITS loading, SkyCalc grid selection, band trimming
`viola_etc/target/photometry.py`	Photometry functions: magnitude conversion, Planck laws, SED construction (Blackbody and PHOENIX-NewEra)
`viola_etc/target/molecules.py`	Molecular template loading, line counting, detection significance estimation
`viola_etc/target/newera_io.py`	PHOENIX-NewEra grid discovery, caching, on-demand download with file locking
`viola_etc/instruments/`	Empty placeholder subpackage reserved for future version (per-instrument configuration modules)
`viola_etc/outputs/`	Empty placeholder subpackage reserved for future version (output formatters/exporters)
`sky_models/`	Precomputed ESO SkyCalc FITS tables, parameterised by altitude and PWV, at $R = 300{,}000$
`molecular_templates/`	Precomputed molecular template CSV files, keyed by planet class, molecule, and band, at $R = 300{,}000$
`stellar_models/`	Precomputed PHOENIX-NewEra stellar atmosphere model grid, parameterised by T_{eff}, at $R = 300{,}000$

2.3 Data Flow Overview

The ETC pipeline follows a physically motivated order. When the user clicks “Run ETC” in the Streamlit UI, the following sequence executes:

Step 1 — Input Assembly. The Streamlit session state values are read and packed into `TargetParams` and `ObservingCondition` dataclasses, which together form the `UserInputs` object. An `InstrumentConfig` and `SiteConfig` are created with their default values (the telescope aperture can be overridden by the user to 2 m, corresponding to the Wendelstein Observatory 2 m

Fraunhofer Telescope (LMU Munich, Germany), or 8 m, corresponding to one unit telescope of the ESO VLT (Paranal, Chile)).

Step 2 — Validation. All inputs are validated by dedicated functions in `validate.py` to catch invalid or out-of-range values early and raise descriptive errors before any computation proceeds.

Step 3 — Sky Model Loading. The SkyCalc FITS grid is queried for the file closest to the requested (altitude, PWV) pair using an L_1-norm nearest-neighbour search. The FITS table is opened, columns are extracted, and the data is trimmed to the selected band window.

Step 4 — Target SED Construction. The target's apparent magnitude is converted to a monochromatic flux density F_ν using either Vega (2MASS) or AB zero-points. A spectral energy distribution is then built on the wavelength grid: either a Planck blackbody anchored to the band-centre flux, or a PHOENIX-NewEra synthetic spectrum interpolated and scaled to match the observed magnitude.

Step 5 — Signal Calculation. The top-of-atmosphere photon rate is computed from the SED, then attenuated by atmospheric transmission, slit throughput (an erf-based model of seeing vs. slit width), and the total instrument throughput. The result is the detected signal in photoelectrons per resolution element.

Step 6 — Background and Noise. Sky emission, instrument thermal emission (Planck B_ν scaled by effective emissivity), dark current, and read noise are computed per resolution element. If A–B nod subtraction is enabled, all Poisson background variances are doubled. Read noise variance is not affected by the nod mode.

Step 7 — SNR Computation. The SNR per resolution element is computed as

$$\mathrm{SNR} = \frac{S}{\sqrt{S + V_{\mathrm{bg}} + V_{\mathrm{read}}}},$$

where S is the signal, V_{bg} is the total Poisson background variance, and V_{read} is the read-noise variance. A magnitude sweep repeats this for a grid of magnitudes while holding backgrounds fixed.

Step 8 — Molecular Detection (optional). If enabled, molecular template CSVs are loaded, resampled onto the ETC wavelength grid, and normalised. Spectral lines are identified via peak finding (SciPy or fallback), filtered by local SNR, and the number of usable lines feeds into a detection significance estimate and required number of observing nights.

Step 9 — Result Packaging. All arrays, summary strings, and debug metadata are packed into an `ETCResult` dataclass and returned to the Streamlit frontend for rendering across five tabs (Results, Sky Model, Signal & Noise, Molecules, Debug). The `viola_etc/` package has no Streamlit dependency in its core modules, so `run_etc()` can also be called directly from a script or Jupyter notebook.

3 Physics Reference

The signal and noise formalism is adapted from Le et al. (2015), modified here for VIOLA instrument parameters. The molecular detection significance is estimated via the high-resolution cross-correlation spectroscopy (HRCCS) framework of Snellen et al. (2015) and Birkby (2018).

3.1 Physical Constants

Table 2: Physical constants used in VIOLA ETC

Symbol	Value	Units	Description
c	2.99792458×10^{8}	$\mathrm{m\,s^{-1}}$	Speed of light
h	$6.62607015 \times 10^{-34}$	$\mathrm{J\,s}$	Planck constant
k_B	1.380649×10^{-23}	$\mathrm{J\,K^{-1}}$	Boltzmann constant

3.2 Magnitude System and Flux Conversion

The ETC supports two photometric systems. In the Vega system, the zero-point flux density $F_{\nu,0}$ is the flux of Vega (here approximated by 2MASS zero-points) in each band. In the AB system, the zero-point is defined as 3631 Jy ($1\,\mathrm{Jy} = 10^{-26}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$) at all frequencies, independent of band.

Table 3: Photometric zero-points

Band	$F_{\nu,0}$ (Vega) [$\mathrm{W\,m^{-2}\,Hz^{-1}}$]	λ range [µm]
J	1.594×10^{-23}	1.10 – 1.30
H	1.024×10^{-23}	1.50 – 1.70
K	6.668×10^{-24}	1.90 – 2.50

The AB zero-point is $F_{\nu,0}(\mathrm{AB}) = 3.631 \times 10^{-23}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$.

Given an apparent magnitude m in either system, the monochromatic flux density is:

$$F_\nu = F_{\nu,0} \cdot 10^{-0.4\,m} \tag{1}$$

3.3 Spectral Energy Distribution

3.3.1 Conversion from F_ν to F_λ

The magnitude gives a monochromatic F_ν at a reference wavelength (here the median of the band). To convert to a spectral flux density per unit wavelength F_λ, we use:

$$F_\lambda\,[\mathrm{W\,m^{-2}\,\mu m^{-1}}] = \frac{c}{\lambda^2} \cdot F_\nu \cdot 10^{-6} \tag{2}$$

where λ is in metres, c/λ^2 converts F_ν [$\mathrm{W\,m^{-2}\,Hz^{-1}}$] to F_λ [$\mathrm{W\,m^{-2}\,m^{-1}}$], and the factor of 10^{-6} converts from per-metre to per-micron. This is evaluated at the reference wavelength $\lambda_{\mathrm{ref}} = \mathrm{median}(\lambda_{\mathrm{grid}})$.

3.3.2 Blackbody SED

The Planck function for spectral radiance per unit wavelength is:

$$B_\lambda(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{\exp\left(\dfrac{hc}{\lambda k_B T}\right) - 1} \tag{3}$$

The result is in $\mathrm{W\,m^{-2}\,sr^{-1}\,m^{-1}}$, then converted to per-micron by multiplying by 10^{-6}.

The stellar SED on the full wavelength grid is obtained by scaling the Planck shape to match the observed flux at the reference wavelength:

$$F_\lambda(\lambda) = F_{\lambda,\mathrm{ref}} \cdot \frac{B_\lambda(\lambda, T_\star)}{B_\lambda(\lambda_{\mathrm{ref}}, T_\star)} \tag{4}$$

This preserves the absolute flux calibration from the magnitude while giving the correct spectral shape for a star of effective temperature $T_\star$.

3.3.3 PHOENIX-NewEra SED

For more realistic spectra, the ETC can use PHOENIX-NewEra synthetic stellar atmosphere models (Hauschildt et al., 2025), selecting models at fixed $\log g = 4.5$ and $[\mathrm{M/H}] = 0.0$ from the available grid. These provide F_λ (in $\mathrm{erg\,s^{-1}\,cm^{-2}\,cm^{-1}}$) as a function of wavelength parameterised by T_{eff}. The grids were downloaded as described in Hauschildt et al. (2025) and convolved with a Gaussian instrumental profile to match the VIOLA resolving power ($R = 300{,}000$) and trimmed to VIOLA wavelength coverage prior to use in the ETC. The unit conversion from CGS surface flux to SI per-micron units is:

$$F_\lambda\,[\mathrm{W\,m^{-2}\,\mu m^{-1}}] = F_\lambda\,[\mathrm{erg\,s^{-1}\,cm^{-2}\,cm^{-1}}] \times 10^{-7} \tag{5}$$

The nearest grid file to the requested T_{eff} is selected, loaded, converted using Equation (5), interpolated onto the ETC wavelength grid, and then scaled so that F_λ at λ_{ref} matches the magnitude-derived value (Equation (2)). The grid covers T_{eff} from 2300 K to 12000 K. Outside this range the code falls back to a blackbody.

3.3.4 Stellar Radial Velocity

A non-zero stellar radial velocity, v_r, shifts the stellar spectrum relative to the telluric rest frame. The observed wavelength is given by the relativistic Doppler relation:

$$\lambda_{\mathrm{obs}} = \lambda_{\mathrm{rest}}\sqrt{\frac{1 + v_r/c}{1 - v_r/c}}, \tag{6}$$

where positive v_r denotes recession (redshift). This displaces stellar absorption lines relative to telluric features and is most relevant for molecular cross-correlation analyses (Section 3.9). For a blackbody SED, the shift displaces a smooth continuum with no spectral features. The current implementation supports v_r in the range $\pm 2000\,\mathrm{km\,s^{-1}}$, covering all known exoplanet host stars. Support for larger values can be extended upon request.

3.4 Observing Mode and Noise Model

3.4.1 Nod-on-Slit Background Subtraction

The ETC supports two observing modes, selectable by the user.

Nod-on-slit mode places the target at two positions along the slit, labelled A and B, separated by a nod throw (e.g., Valenti et al. 2021, Section 5.2.5.1). Both positions remain on the slit, so the stellar signal is recorded in every frame. The telescope alternates between A and B, accumulating N_{exp} frames in total ($N_{\mathrm{exp}}/2$ at each position). During data reduction, each A–B frame pair is subtracted to remove the mean background level (sky emission, thermal emission, and dark current).

The ETC predicts the post-reduction SNR, assuming that sky emission, instrument thermal radiation, and dark current have been removed from the spectrum during data reduction. These background sources nevertheless contribute shot noise. Because A and B are independent exposures, each carrying its own background shot noise, the variance of the subtracted frame is:

$$\mathrm{Var}(A - B) = N_{\mathrm{bg}} + N_{\mathrm{bg}} = 2\,N_{\mathrm{bg}}$$

where $N_{\mathrm{bg}} = N_{\mathrm{sky}} + N_{\mathrm{th}} + N_{\mathrm{dark}}$ is the total background count per frame, assumed equal at both nod positions. Background removal eliminates the mean level but retains this doubled Poisson variance. This noise penalty is captured by a *pair factor* $f_{\mathrm{pair}} = 2$.

No-nodding mode ($f_{\mathrm{pair}} = 1$) assumes that background is estimated and removed without incurring an additional shot-noise penalty (optimistic case). The background Poisson variance is counted once.

3.4.2 Noise Sources

The total noise variance per spectral resolution element comprises three independent contributions. Notation used throughout: S_{res} [e^-] is the detected stellar signal per resolution element (Section 3.5); N_{exp} is the total number of frames; t_{exp} [s] is the exposure time per frame; $t_{\mathrm{total}} = t_{\mathrm{exp}} \times N_{\mathrm{exp}}$ [s] is the total integration time; and $n_{\mathrm{pix}} = n_x \times n_y$ is the number of detector pixels per resolution element.

(1) Signal shot noise. Photon arrivals obey Poisson statistics, so the variance equals the mean detected signal:

$$V_{\mathrm{sig}} = S_{\mathrm{res}}$$

(2) Background shot noise. Three background sources contribute Poisson variance:

- N_{sky}: sky emission (OH airglow, thermal continuum, and other atmospheric emission) transmitted through the instrument optics;
- N_{th}: instrument thermal emission from the telescope and warm optical components, modelled as a Planck function B_ν scaled by the effective emissivity $\varepsilon_{\mathrm{eff}}$;
- N_{dark}: detector dark current, $N_{\mathrm{dark}} = i_{\mathrm{dark}}\, t_{\mathrm{total}}\, n_{\mathrm{pix}}$, where i_{dark} [$\mathrm{e}^-\,\mathrm{s}^{-1}\,\mathrm{pix}^{-1}$] is the dark current rate.

All three are subtracted in mean during data reduction, but their Poisson fluctuations remain and are scaled by the pair factor:

$$V_{\mathrm{bg}} = f_{\mathrm{pair}}\, N_{\mathrm{bg}} = f_{\mathrm{pair}}\,(N_{\mathrm{sky}} + N_{\mathrm{th}} + N_{\mathrm{dark}})$$

(3) Read noise. Each detector readout introduces a fixed Gaussian noise σ_{RN} per pixel, independent of the signal level. The total read-noise variance over N_{exp} frames and n_{pix} pixels per resolution element is:

$$V_{\mathrm{read}} = \sigma_{\mathrm{RN}}^2\, N_{\mathrm{exp}}\, n_{\mathrm{pix}}$$

The pair factor does not apply to read noise. In nod-on-slit mode, all N_{exp} frames are read ($N_{\mathrm{exp}}/2$ at each nod position), each contributing one read independently.

3.5 Signal Calculation

The detected signal per resolution element is computed step by step through the signal path described below.

3.5.1 Photon Rate at Top of Atmosphere

The energy of a single photon at wavelength λ is:

$$E_{\rm ph}(\lambda) = \frac{hc}{\lambda} \tag{7}$$

The photon arrival rate per unit collecting area and per unit wavelength interval is:

$$\Phi_{\rm TOA}(\lambda) = \frac{F_\lambda(\lambda)}{E_{\rm ph}(\lambda)} \quad [\mathrm{ph\,s^{-1}\,m^{-2}\,\mu m^{-1}}] \tag{8}$$

3.5.2 Atmospheric Transmission

The source photon rate is attenuated by the wavelength-dependent telluric transmission $\tau_{\rm atm}(\lambda)$, obtained from precomputed ESO SkyCalc models (see Section 3.6 for details). The ground-level photon rate is:

$$\Phi_{\rm ground}(\lambda) = \Phi_{\rm TOA}(\lambda) \cdot \tau_{\rm atm}(\lambda) \quad [\mathrm{ph\,s^{-1}\,m^{-2}\,\mu m^{-1}}] \tag{9}$$

3.5.3 Photons Collected by the Telescope

For a spectrograph with resolving power $R = \lambda/\Delta\lambda$, the bandwidth per resolution element is:

$$\Delta\lambda(\lambda) = \frac{\lambda}{R} \tag{10}$$

The number of photons per resolution element collected over the total integration time is:

$$S_{\rm ground} = \Phi_{\rm ground} \cdot \Delta\lambda \cdot A_{\rm tel} \cdot t_{\rm total} \tag{11}$$

where $A_{\rm tel} = \pi(D/2)^2$ is the telescope collecting area and $t_{\rm total} = t_{\rm exp} \times N_{\rm exp}$.

3.5.4 Slit Throughput

The fraction of light from a point source that passes through the spectrograph slit depends on the seeing FWHM relative to the slit width. For a Gaussian seeing profile with FWHM $= \theta_{\rm see}$, the fraction of light falling within a slit of width w is:

$$\tau_{\rm slit} = \mathrm{erf}\left(\sqrt{\ln 2} \cdot \frac{w}{\theta_{\rm see}}\right) \tag{12}$$

This comes from integrating the 1D Gaussian profile across the slit.

3.5.5 Instrument Parameters

Table 4 summarises all fixed instrument parameters used by the ETC. The total instrument throughput is the product of all component transmissions:

$$\tau_{\rm inst} = \tau_{M1} \cdot \tau_{\rm window} \cdot \tau_{\rm FRD} \cdot \tau_{\rm doublet} \cdot \tau_{\rm collimator} \cdot \tau_{\rm vipa} \cdot \tau_{\rm xdisp} \cdot \tau_{\rm camera} \cdot \tau_{\rm detector} \tag{13}$$

Table 4: VIOLA instrument parameters (ETC defaults)

Parameter	**Value**	**Description**
Telescope and Spectrograph		
D	2.0 m (default)	Telescope diameter; configurable to 8 m (VLT)
$A_{\mathrm{tel}} = \pi(D/2)^2$	3.14 m^2	Collecting area at $D = 2\,\mathrm{m}$
R	300 000	Spectral resolving power
Bands	J: 1.10–1.30 µm	
	H: 1.50–1.70 µm	
	K: 1.90–2.50 µm	
Slit and Pixel Geometry		
w	1.6 arcsec	Slit width
s_{pix}	0.8 arcsec pix^{-1}	Spatial plate scale
n_x	2 pix	Pixels per res el (dispersion)
n_y	2 pix	Pixels per res el (spatial)
$n_{\mathrm{pix}} = n_x \times n_y$	4 pix	Total pixels per resolution element
Detector		
σ_{RN}	20 e^- pix^{-1}	Read noise per pixel per read
i_{dark}	0.01 e^- s^{-1} pix^{-1}	Dark current rate
Thermal Background		
T_{amb}	275 K	Instrument ambient temperature
$\varepsilon_{\mathrm{eff}}$	0.10	Effective emissivity (scales Planck B_ν)
Optical Throughput		
τ_{M1}	0.95	Primary mirror reflectivity
τ_{window}	0.95	Entrance window
τ_{FRD}	0.90	Fiber focal ratio degradation
τ_{doublet}	0.96	Doublet lens
$\tau_{\mathrm{collimator}}$	0.90	Collimator optics
τ_{vipa}	0.90	VIPA grating efficiency
τ_{xdisp}	0.80	Cross-disperser
τ_{camera}	0.90	Camera optics
τ_{detector}	0.95	Detector quantum efficiency
τ_{inst}	≈ 0.43	Product of all components above

3.5.6 Final Signal per Resolution Element

Combining all factors, the detected signal in photoelectrons per resolution element is:

$$S_{\mathrm{res}} = S_{\mathrm{ground}} \cdot \tau_{\mathrm{inst}} \cdot \tau_{\mathrm{slit}} \quad [\mathrm{e}^{-}/\mathrm{res}] \tag{14}$$

The corresponding per-pixel signal is $S_{\mathrm{pix}} = S_{\mathrm{res}}/n_{\mathrm{pix}}$, where $n_{\mathrm{pix}} = n_x \times n_y$ is the number of detector pixels per resolution element (n_x and n_y in the dispersion and spatial directions, respectively).

3.6 Background Sources

3.6.1 Extraction Solid Angle

The solid angle subtended by one resolution element on the sky is determined by the slit width and the spatial extent of the extraction aperture:

$$\Omega_{\mathrm{res}} = w_{\mathrm{slit}} \cdot (s_{\mathrm{pix}} \cdot n_y) \quad [\mathrm{arcsec}^2] \tag{15}$$

where w_{slit} is the slit width, s_{pix} the spatial plate scale, and n_y the number of extracted rows (see Table 4 for values). For thermal calculations, this is converted to steradians: $\Omega_{\mathrm{sr}} = \Omega_{\mathrm{res}} \cdot (\pi/648000)^2$.

3.6.2 Sky Emission Background

The sky background is modelled using the ESO SkyCalc tool (Noll et al., 2012; Jones et al., 2013), calibrated for ESO Cerro Paranal. It provides a good approximation for VLT observations. For Wendelstein Observatory it serves as a working substitute, as the actual sky conditions there may differ considerably from Paranal.

Each SkyCalc model file delivers two sets of spectra. The first is the atmospheric transmission $\tau_{\mathrm{atm}}(\lambda)$, decomposed into four components (molecular absorption, ozone, Rayleigh scattering, and Mie scattering). The total atmospheric transmission is used in the signal calculation (Equation (9)). The second is the sky emission spectrum, decomposed into seven components, each in units of $\mathrm{ph\,s^{-1}\,m^{-2}\,\mu m^{-1}\,arcsec^{-2}}$. The ETC sums six of them:

$$\Phi_{\mathrm{sky}} = \Phi_{\mathrm{ZL}} + \Phi_{\mathrm{AEL}} + \Phi_{\mathrm{TME}} + \Phi_{\mathrm{SML}} + \Phi_{\mathrm{SSL}} + \Phi_{\mathrm{ARC}} \tag{16}$$

- Φ_{ZL} — zodiacal light (scattered sunlight from interplanetary dust)
- Φ_{AEL} — airglow emission lines (OH and other upper-atmosphere species)
- Φ_{TME} — molecular thermal emission from the lower atmosphere
- Φ_{SML} — scattered moonlight
- Φ_{SSL} — scattered starlight
- Φ_{ARC} — airglow residual continuum

The seventh component, Φ_{TIE} (telescope/instrument thermal emission), is excluded from the sum. The ETC models instrument thermal emission independently via a Planck greybody (Section 3.6.3).

The ETC draws these spectra from a precomputed grid parameterised by target altitude and precipitable water vapour (PWV). The user may specify any value within the covered ranges (30°–90° for altitude; 0.5–20.0 mm for PWV), and the nearest grid node is selected automatically. The grid nodes are:

$$\mathrm{Altitude} \in \{30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ\}$$
$$\mathrm{PWV} \in \{0.5, 1.0, 1.5, 2.5, 3.5, 5.0, 7.5, 10.0, 20.0\}\ \mathrm{mm}$$

All other SkyCalc parameters are held fixed at the values listed in Table A1 in Appendix A. The dependence of each sky component on altitude and PWV is illustrated in Appendix A. A future release will support user-supplied sky models.

The sky background in electrons per resolution element is:

$$N_{\rm sky} = A_{\rm tel} \cdot \Omega_{\rm res} \cdot \tau_{\rm inst} \cdot \Delta\lambda \cdot \Phi_{\rm sky} \cdot t_{\rm total} \tag{17}$$

Only $\tau_{\rm inst}$ is applied, not $\tau_{\rm atm}$. SkyCalc already delivers ground-level brightness with atmospheric transmission folded in. No slit-loss factor is applied because sky emission is an extended source. Here, $\Omega_{\rm res}$ is the extraction solid angle defined in Equation (15).

3.6.3 Instrument Thermal Background

The telescope mirrors and instrument optics emit thermal radiation. This is modelled as a greybody: a Planck (blackbody) spectrum B_ν at the ambient temperature $T_{\rm amb}$ scaled by an effective emissivity $\varepsilon_{\rm eff}$, which represents the combined emissivity of all optical surfaces. The adopted values are listed in Table 4:

$$B_\nu(\nu, T) = \frac{2h\nu^3}{c^2} \cdot \frac{1}{\exp\left(\frac{h\nu}{k_B T}\right) - 1} \quad [\mathrm{W\,m^{-2}\,sr^{-1}\,Hz^{-1}}] \tag{18}$$

The thermal background rate per resolution element is:

$$R_{\rm thermal} = \frac{A_{\rm tel} \cdot \Omega_{\rm sr}}{h \cdot R} \cdot \varepsilon_{\rm eff} \cdot B_\nu \tag{19}$$

The factor $1/(hR)$ combines the bandwidth of one resolution element ($\Delta\nu = \nu/R$) and the photon energy ($h\nu$); the ν cancels, leaving a frequency-independent conversion from power per Hz to photon rate per resolution element. The total thermal contribution is $N_{\rm th} = R_{\rm thermal} \cdot t_{\rm total}$.

Note that $R_{\rm thermal}$ accounts for instrument emission only. The thermal emission from the atmosphere itself ($\Phi_{\rm TME}$ in Equation (16)) is already included in the sky background term via the SkyCalc model.

3.6.4 Detector Noise

Two detector-related noise sources are included.

Dark current: $N_{\rm dark} = i_{\rm dark} \cdot t_{\rm total} \cdot n_{\rm pix}$, where $i_{\rm dark}$ is the dark current rate and $n_{\rm pix} = n_x \cdot n_y$ is the number of pixels per resolution element (see Table 4 for adopted values).

Read noise: The read-noise variance per resolution element for $N_{\rm exp}$ frames is:

$$V_{\rm read} = \sigma_{\rm RN}^2 \cdot N_{\rm exp} \cdot n_{\rm pix} \tag{20}$$

where $\sigma_{\rm RN}$ is the read noise per pixel per frame (see Table 4).

3.7 SNR Calculation

The SNR per spectral resolution element is:

$$\mathrm{SNR} = \frac{S_{\rm res}}{\sqrt{S_{\rm res} + V_{\rm bg} + V_{\rm read}}} \tag{21}$$

where the total Poisson background variance is:

$$V_{\rm bg} = f_{\rm pair} \cdot (N_{\rm sky} + N_{\rm th} + N_{\rm dark}) \tag{22}$$

where $f_{\rm pair}$ is the nod-subtraction pair factor defined in Section 3.4.1. The read-noise variance $V_{\rm read}$ (Equation (20)) is not multiplied by $f_{\rm pair}$.

3.8 SNR vs. Magnitude Sweep

To produce the diagnostic plot of SNR as a function of target magnitude, the ETC assumes that background terms (sky, thermal, dark, read noise) are independent of the target brightness. The signal is rescaled using the relation:

$$S_{\rm res}(m) = S_{\rm res}(m_0) \cdot 10^{-0.4\,(m-m_0)} \tag{23}$$

where m_0 is the user-specified magnitude. For each magnitude in the sweep grid, the SNR is recomputed using Equation (21) with the rescaled signal, and the median across all wavelengths is reported.

3.9 Molecular Detection Estimation

The ETC provides a first-order estimate of the molecular detection significance achievable in one allocated observing night, and the number of observing nights required to reach a user-specified detection significance.

The calculation converts the stellar signal and noise budget from Sections 3.5–3.7 into an approximate high-resolution cross-correlation spectroscopy (HRCCS) detectability metric. This estimate is a forecasting tool, not a replacement for a full HRCCS analysis. It depends on the selected planet-class template described in Appendix B and on simplified assumptions about line contrast, line independence, observing stability, and signal survival through detrending.

3.9.1 Detection Significance per Observing Night

The HRCCS detection significance for one observing night is built from $\mathrm{SNR}_{\mathrm{res},i}$ (Equation (21)), the post-reduction SNR per resolution element accumulated over the full night. Because $S_{\rm res} \propto N_{\rm exp}\, t_{\rm exp}$ already integrates over all frames and $V_{\rm bg}$ carries the nod-subtraction pair factor $f_{\rm pair}$ (Section 3.4.1), no additional frame-stacking factor is applied.

Template preparation. Before lines are selected, the molecular template (Appendix B) is Doppler-shifted by the stellar systemic velocity $v_{\rm rv}$ (`v_rv_km_s`) using Equation (6), placing the planet lines at their observer-frame wavelengths under the assumption that the planet atmosphere co-moves with the star. Exposure-by-exposure orbital Doppler shifts and planet atmosphere dynamics are not modelled. Their effect is captured by the user-supplied detrending factor $p_{\rm detrend}$ described below.

Line selection. A candidate line at λ_j is counted as usable ($j \in \mathcal{L}$, $N_{\rm lines} = |\mathcal{L}|$) only if it satisfies both:

1. its template prominence exceeds the user-supplied threshold `prom_abs` (see Appendix B);
2. $\mathrm{SNR}_{\rm res}(\lambda_j) \geq \mathrm{SNR}_{\rm thr}$ (user-supplied `snr_thresh`), removing lines that fall in deep telluric absorption regions or continuum-suppressed wavelengths.

A minimum of `min_lines_for_calc` usable lines is required before the calculation is attempted. Otherwise, both $\mathrm{SNR}_{\rm night}$ and $N_{\rm nights}$ are reported as N/A.

Per-line detection SNR. Each usable line is treated following the matched-filter framework derived in Appendix C. The planet imprints a fractional signal $C_{\rm line}$ (the planet-to-star line contrast) on the stellar continuum, so the raw planet signal at line j is

$$S_{\mathrm{p},j} = C_{\rm line}\, S_{\mathrm{res},j}. \tag{24}$$

After passing through the detrending step of the data-reduction pipeline, only a fraction $p_{\rm detrend}$ of this signal survives:

$$S_{{\rm p,rec},j} = p_{\rm detrend}\, C_{\rm line}\, S_{{\rm res},j}, \tag{25}$$

giving a per-line detection SNR of

$$\mathrm{SNR}_{{\rm line},j} = p_{\rm detrend}\, C_{\rm line}\, \mathrm{SNR}_{{\rm res},j}. \tag{26}$$

Combining lines. Treating the usable lines as independent noise channels (Appendix C, Equation (C10)), the per-line SNRs add in quadrature to give the total detection significance for one night:

$$\mathrm{SNR}_{\rm night} \simeq p_{\rm detrend}\, C_{\rm line} \left[\sum_{j\in\mathcal{L}} \mathrm{SNR}^2_{{\rm res},j}\right]^{1/2}. \tag{27}$$

In the uniform-SNR limit, where $\mathrm{SNR}_{{\rm res},j} \approx \mathrm{SNR}_\star$ for all lines, this reduces to the familiar HRCCS scaling (Snellen et al., 2015; Birkby, 2018):

$$\mathrm{SNR}_{\rm CCF} \simeq p_{\rm detrend}\, C_{\rm line}\, \mathrm{SNR}_\star \sqrt{N_{\rm lines}}. \tag{28}$$

The detrending factor $p_{\rm detrend}$. The factor $p_{\rm detrend}$ (`detrend_p`) corresponds to the signal-throughput factor α in Appendix C. It represents the fraction of the planet signal that survives telluric removal, stellar-line subtraction, wavelength-alignment residuals, and other pipeline losses. Users may adopt a smaller value for conservative forecasts, or set it to a measured injection-recovery fraction from their own pipeline.

3.9.2 Required Number of Observing Nights

Assuming independent observing nights with comparable observing conditions, the detection significance grows as $\sqrt{N_{\rm nights}}$. The number of allocated observing nights required to reach a target detection significance $\sigma_{\rm det}$ is

$$N_{\rm nights} = \left(\frac{\sigma_{\rm det}}{\mathrm{SNR}_{\rm night}}\right)^2. \tag{29}$$

Here $\sigma_{\rm det}$ is the user-specified target detection significance, for example 5σ.

4 Data Models Reference

4.1 TargetParams

Table 5: `TargetParams` fields

Field	Type	Default	Description
`band`	`str`	—	Observing band: `"J"`, `"H"`, or `"K"`
`mag_system`	`str`	—	Magnitude system: `"Vega"` or `"AB"`
`m_mag`	`float`	—	Apparent magnitude in the selected band
`source_sed`	`str`	`"blackbody"`	SED model: `"blackbody"` or `"phoenix-newera"`
`T_star_K`	`float`	5800.0	Stellar effective temperature [K]
`v_rv_km_s`	`float`	0.0	Stellar radial velocity [$\mathrm{km\,s^{-1}}$]; shifts PHOENIX-NewEra spectral lines relative to tellurics and adjusts the blackbody continuum shape via rest-frame wavelength evaluation. Range: $\pm 2000\,\mathrm{km\,s^{-1}}$. See Section 3.3.4.
`planet_line_contrast`	`float`	10^{-4}	Planet-to-star line contrast $C_{\rm line}$

4.2 UserInputs

`UserInputs` is the top-level input container passed to `run_etc()`. It groups the two user-facing parameter dataclasses and has no fields of its own.

Table 6: `UserInputs` fields

Field	Type	Description
`target`	`TargetParams`	Target star / planet signal parameters (see above)
`obs`	`ObservingCondition`	Observing conditions and exposure settings (see below)

4.3 InstrumentConfig

Table 7: `InstrumentConfig` fields

Field	Type	Default	Description
`d_m`	`float`	2.0	Telescope diameter [m]
`resolving_power`	`float`	300,000	Spectral resolving power R
`nx`	`int`	2	Pixels per res element (dispersion)
`ny`	`int`	2	Pixels per res element (spatial)
`band_edges_um`	`dict`	J/H/K windows	Per-band wavelength windows [µm]: J $(1.10, 1.30)$, H $(1.50, 1.70)$, K $(1.90, 2.50)$
`slit_width_as`	`float`	1.6	Slit width [arcsec]
`pix_scale_ny`	`float`	0.8	Spatial pixel scale [arcsec pix^{-1}]
`optics`	`dict`	9 factors	Per-element optical throughput factors (M1, window, FRD, doublet, collimator, vipa, cross-disperser, camera, detector); product gives $\tau_{\rm inst}$
`t_ambient_k`	`float`	275.0	Ambient instrument temperature [K]
`epsilon_eff`	`float`	0.10	Effective thermal emissivity
`rn_e`	`float`	20.0	Read noise $\sigma_{\rm RN}$ [e$^-$ pix^{-1} read^{-1}]
`dark_rate`	`float`	0.01	Dark current [e$^-$ s^{-1} pix^{-1}]

4.4 ObservingCondition

Table 8: `ObservingCondition` fields

Field	Type	Default	Description
`t_exp_s`	`float`	—	Exposure time per frame [s]
`n_exp`	`int`	—	Total number of frames in one observing night
`use_nod_subtraction`	`bool`	—	Enable A–B nod subtraction
`seeing_fwhm_as`	`float`	—	Seeing FWHM [arcsec]
`target_alt_deg`	`float`	60.0	Target altitude [deg]
`pwv_mm`	`float`	1.0	Precipitable water vapour [mm]
`sky_model_name`	`str`	`"SkyCalc_Grid"`	Sky-model dispatch key; selects the 45-file SkyCalc grid loader

4.5 SiteConfig

Table 9: `SiteConfig` fields

Field	Type	Default	Description
`sky_model_base_dir`	`str`	`"./sky_models"`	Directory containing the SkyCalc FITS grid

4.6 ETCResult

The output container returned by `run_etc()`. All array fields are `None` if not computed. Molecule fields default to empty dicts to prevent `AttributeError` in the UI.

Table 10: `ETCResult` output fields

Field	Type	Description
`lam_um`	ndarray	Wavelength grid [µm], trimmed to the selected band
`trans`	ndarray	Atmospheric transmission spectrum; also accessible via `sky.trans`
`sky`	`SkyData`	All sky-model arrays on the band-trimmed wavelength grid; see Table 11
`signal_res_e`	ndarray	Signal per resolution element [e^-]
`signal_pix_e`	ndarray	Signal per pixel [e^-]
`sky_res_e`	ndarray	Sky background per resolution element [e^-]
`thermal_res_e`	ndarray	Instrument thermal background per resolution element [e^-]
`dark_res_e`	ndarray	Dark current per resolution element [e^-]
`snr_res`	ndarray	SNR per resolution element
`mag_grid`	ndarray	Magnitude sweep grid
`snr_med_grid`	ndarray	Median SNR per resolution element at each magnitude
`molecule_templates`	dict	Loaded molecular templates, keyed by molecule name
`molecule_metrics`	dict	Detection significance and required observing nights per molecule
`summary_lines`	list[str]	Human-readable summary of key result scalars
`meta`	dict	All scalar diagnostics (inputs, throughputs, medians, sweep range)

Table 11: `SkyData` fields (all arrays on the band-trimmed wavelength grid)

Field	Source column	Description
`lam_um`	`lam`	Vacuum wavelength [µm] (converted from nm)
`trans`	`trans`	Total atmospheric transmission [0–1]
`sky_phi`	*derived*	Sum of the six emission components below (`flux_tie` excluded) [$\mathrm{ph\,s^{-1}\,m^{-2}\,\mu m^{-1}\,arcsec^{-2}}$]
`flux_zl`	`flux_zl`	Zodiacal light
`flux_ael`	`flux_ael`	Airglow emission lines (upper atmosphere)
`flux_tme`	`flux_tme`	Molecular emission of the lower atmosphere
`flux_sml`	`flux_sml`	Scattered moonlight
`flux_ssl`	`flux_ssl`	Scattered starlight
`flux_arc`	`flux_arc`	Airglow residual continuum
`flux_tie`	`flux_tie`	SkyCalc telescope/instrument thermal (carried for diagnostics only; the ETC computes its own thermal background via a Planck greybody and excludes `flux_tie` from `sky_phi`)
`trans_ma`	`trans_ma`	Molecular absorption transmission
`trans_o3`	`trans_o3`	Ozone absorption transmission
`trans_rs`	`trans_rs`	Rayleigh scattering transmission
`trans_ms`	`trans_ms`	Mie scattering transmission

4.7 `run_etc()` API Reference

The main entry point is `run_etc(u, cfg, site, ...)` in `viola_etc/core/runner.py`. The required argument is `u` (`UserInputs`); all others have defaults.

Table 12: `run_etc()` optional parameters

Parameter	Default	Description
`cfg`	`InstrumentConfig()`	Instrument configuration; override to change telescope diameter or detector parameters
`site`	`SiteConfig()`	Site configuration; sets the sky model base directory
`mag_sweep_min`	5.0	Minimum magnitude for SNR sweep
`mag_sweep_max`	16.0	Maximum magnitude for SNR sweep
`mag_sweep_step`	0.5	Step size for SNR sweep
`molecule_dir`	`"./molecular_templates"`	Path to molecular template CSV files
`enable_molecules`	`True`	Run the molecular detection module
`class_name`	`None`	Planet-class selector for the v0.5 template grid (e.g. `"temperate_terrestrial"`, `"hot_jupiter"`); when set, only templates with the matching filename prefix are loaded
`detect_sig`	5.0	Required detection significance [σ]
`prom_abs`	0.05	Minimum line depth (normalised) for line selection
`snr_thresh`	100.0	Minimum SNR per resolution element for line selection
`detrend_p`	0.5	Detrending efficiency (0–1)
`min_lines_for_calc`	50	Minimum number of lines required to compute detection significance

5 Version History

Table 13: Version history

Version	Key Changes
v0.1	Initial Python implementation. Core signal and noise calculation. Command-line interface only.
v0.2	Refined signal and noise calculation pipeline. Improved noise model and molecular detection framework. Command-line interface only.
v0.3	First Streamlit web application. Interactive UI for target, observing conditions, and SNR output.
v0.4	Added PHOENIX-NewEra SED model. SkyCalc grid with altitude/PWV selection. Configurable telescope aperture (2 m / 8 m).
v0.5 (current)	Corrected sky background calculation to properly use ESO SkyCalc ground-level brightness. Added stellar radial velocity shift (`v_rv_km_s`) for the stellar spectrum and planet template. Updated detector throughput from 0.80 to 0.95. Upgraded molecular detection to a precomputed five-class planet template grid. Bug fixes: read-noise variance no longer multiplied by the pair factor; N_{exp} double-counting removed from molecular SNR formula; detrending factor p_{detrend} moved outside the square root.

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A SkyCalc Sky Model Grid

A.1 Grid Overview

The ETC uses a precomputed grid of 45 ESO SkyCalc FITS files generated with the `skycalc_cli` command-line tool (SkyCalc v2.0.9).[1] The grid is parameterised by two axes:

Axis	Values
Altitude (airmass X)	90°, 75°, 60°, 45°, 30° ($X = 1.0, 1.035, 1.155, 1.414, 2.0$)
PWV	0.5, 1.0, 1.5, 2.5, 3.5, 5.0, 7.5, 10.0, 20.0 mm

All remaining parameters are fixed and chosen to represent moderate, typical observing conditions. Table A1 lists the complete set.

[1] https://www.eso.org/observing/etc/doc/skycalc/helpskycalccli.html

Table A1: Fixed SkyCalc parameters common to all 45 grid files. Values represent moderate / typical conditions unless stated.

Parameter	**Value**	**Description / scenario rationale**
Site and wavelength grid		
`observatory`	`paranal`	ESO Cerro Paranal (2640 m as used by SkyCalc), Chile
`vacair`	`vac`	Output wavelengths in vacuum
`wmin`/`wmax`	900–2750 nm	Covers full J, H, and K bands
`wres`	300 000	Spectral resolving power R
Solar activity		
`msolflux`	130 SFU	Moderate solar activity; SkyCalc default
Moon (moderate lunar conditions)		
`moon_sun_sep`	90°	SkyCalc default
`moon_target_sep`	45°	SkyCalc default
`moon_alt`	45°	SkyCalc default
`moon_earth_dist`	1.0	SkyCalc default
Zodiacal and scattered starlight		
`incl_zodiacal`	`Y`	Zodiacal scattered sunlight included (`flux_zl`)
`ecl_lat`	90°	SkyCalc default (ecliptic pole; minimises zodiac light, optimistic scenario)
`ecl_lon`	135°	SkyCalc default
`incl_starlight`	`Y`	Scattered starlight included (`flux_ssl`)
Atmospheric emission components		
`incl_loweratm`	`Y`	Lower-atmosphere molecular emission included (`flux_tme`)
`incl_upperatm`	`Y`	Upper-atmosphere airglow lines included (`flux_ael`)
`incl_airglow`	`Y`	Airglow/residual continuum included (`flux_arc`)
`incl_therm`	`Y`	Telescope/instrument thermal included (`flux_tie`)
Season and time		
`season`	0	Annual mean (no specific season selected)
`time`	0	Annual mean (no specific time of night selected)

Each FITS binary table provides 14 columns. Transmission columns: `trans` (total), `trans_ma` (molecular absorption), `trans_o3` (ozone), `trans_rs` (Rayleigh scattering), `trans_ms` (Mie scattering). Emission columns (ph $\mathrm{s}^{-1}\,\mathrm{m}^{-2}\,\mu\mathrm{m}^{-1}\,\mathrm{arcsec}^{-2}$): `flux` (total), `flux_ael` (airglow lines), `flux_tme` (molecular emission, lower atmosphere), `flux_sml` (scattered moonlight), `flux_ssl` (scattered starlight), `flux_zl` (zodiacal light), `flux_arc` (airglow/residual continuum), `flux_tie` (telescope/instrument thermal).

A.2 Band-Averaged Statistics

Table A2 lists the mean transmission and median total sky emission flux in each of the three VIOLA science bands for all 45 grid points.

Table A2: Band-averaged sky model statistics for all 45 SkyCalc grid points.

Alt (deg)	PWV (mm)	Mean transmission [0–1] J	H	K	Median flux ($\mathrm{ph\,s^{-1}\,m^{-2}\,\mu m^{-1}\,arcsec^{-2}}$) J	H	K
90	0.5	0.954	0.982	0.875	344	683	8,573
90	1.0	0.940	0.981	0.854	334	683	8,635
90	1.5	0.930	0.980	0.839	321	683	8,670
90	2.5	0.913	0.978	0.818	309	682	8,746
90	3.5	0.899	0.976	0.801	299	681	8,795
90	5.0	0.881	0.974	0.782	296	679	8,879
90	7.5	0.857	0.971	0.758	289	677	9,005
90	10.0	0.838	0.968	0.740	283	676	9,212
90	20.0	0.781	0.957	0.686	270	669	9,943
75	0.5	0.953	0.981	0.872	354	705	8,581
75	1.0	0.939	0.980	0.851	343	705	8,637
75	1.5	0.928	0.979	0.836	330	704	8,679
75	2.5	0.910	0.977	0.814	317	704	8,751
75	3.5	0.896	0.976	0.798	308	703	8,800
75	5.0	0.879	0.973	0.779	304	701	8,884
75	7.5	0.855	0.970	0.755	297	699	9,024
75	10.0	0.835	0.967	0.736	290	698	9,223
75	20.0	0.777	0.956	0.682	278	690	9,990
60	0.5	0.948	0.979	0.864	387	779	8,617
60	1.0	0.934	0.978	0.842	377	779	8,657
60	1.5	0.922	0.977	0.827	361	779	8,723
60	2.5	0.904	0.975	0.804	346	777	8,772
60	3.5	0.889	0.973	0.787	338	776	8,819
60	5.0	0.870	0.971	0.768	333	775	8,915
60	7.5	0.845	0.967	0.743	324	773	9,100
60	10.0	0.825	0.964	0.724	316	771	9,276
60	20.0	0.765	0.952	0.670	304	759	10,108
45	0.5	0.939	0.975	0.847	460	936	8,757
45	1.0	0.923	0.974	0.824	442	936	8,806
45	1.5	0.910	0.972	0.808	425	935	8,872
45	2.5	0.890	0.970	0.784	405	934	8,936
45	3.5	0.874	0.968	0.767	399	933	9,023
45	5.0	0.853	0.965	0.747	392	930	9,209
45	7.5	0.826	0.961	0.721	380	927	9,511
45	10.0	0.804	0.957	0.702	371	923	9,799
45	20.0	0.742	0.944	0.645	359	909	10,519
30	0.5	0.921	0.966	0.815	613	1,271	8,919
30	1.0	0.901	0.964	0.789	576	1,270	8,975
30	1.5	0.886	0.963	0.772	556	1,268	9,069
30	2.5	0.862	0.959	0.747	538	1,264	9,237
30	3.5	0.844	0.957	0.728	529	1,262	9,426
30	5.0	0.820	0.953	0.707	515	1,259	9,705
30	7.5	0.790	0.948	0.681	498	1,250	10,136
30	10.0	0.765	0.943	0.660	489	1,243	10,355

(continued)

Alt (deg)	PWV (mm)	Mean transmission [0–1] J	H	K	Median flux ($\mathrm{ph\,s^{-1}\,m^{-2}\,\mu m^{-1}\,arcsec^{-2}}$) J	H	K
30	20.0	0.698	0.928	0.601	477	1,220	10,953

A.3 Transmission Components

Figure A1 shows the mean band transmission for each of the five transmission components as a function of altitude and PWV. Figures A2 and A3 show full spectra varying PWV at fixed altitude and varying altitude at fixed PWV, respectively. Three of the five components remain very close to unity across all J, H, and K bands: `trans_o3` (ozone), `trans_rs` (Rayleigh scattering), and `trans_ms` (Mie scattering). Ozone absorption is concentrated in the UV and visible, becoming negligible beyond $\sim$1 µm (`trans_o3` ≈ 1.000). Both Rayleigh and aerosol (Mie) scattering weaken at longer wavelengths. The dominant source of transmission loss in the NIR is therefore molecular absorption (`trans_ma`).

A.4 Emission Components

Figures A4 and A5 show the median band flux for each emission component across the 5×9 altitude–PWV grid. Figures A6 and A7 show full spectra varying PWV and altitude respectively. The `flux_ael` (airglow emission lines) panels show zero in all bands because airglow lines are narrow features that occupy only a small fraction of the total wavelength pixels within each band. The median over all pixels in the band is therefore zero even though the lines themselves can be bright. The line contribution is visible in the spectra (Figures A6 and A7).

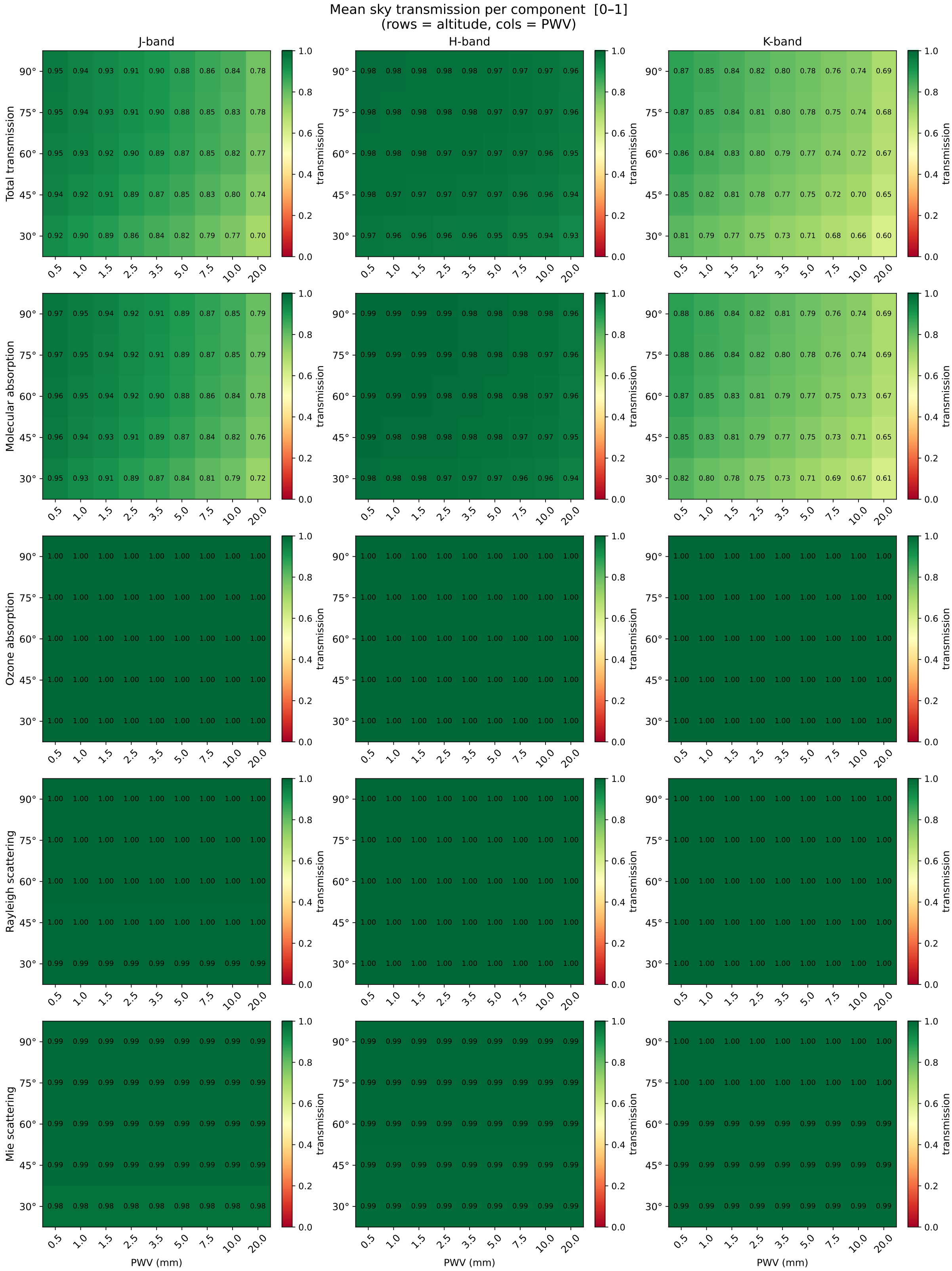

Figure A1: Mean transmission per component and VIOLA band across the 5×9 altitude–PWV grid. Rows = altitude (deg); columns = PWV (mm).

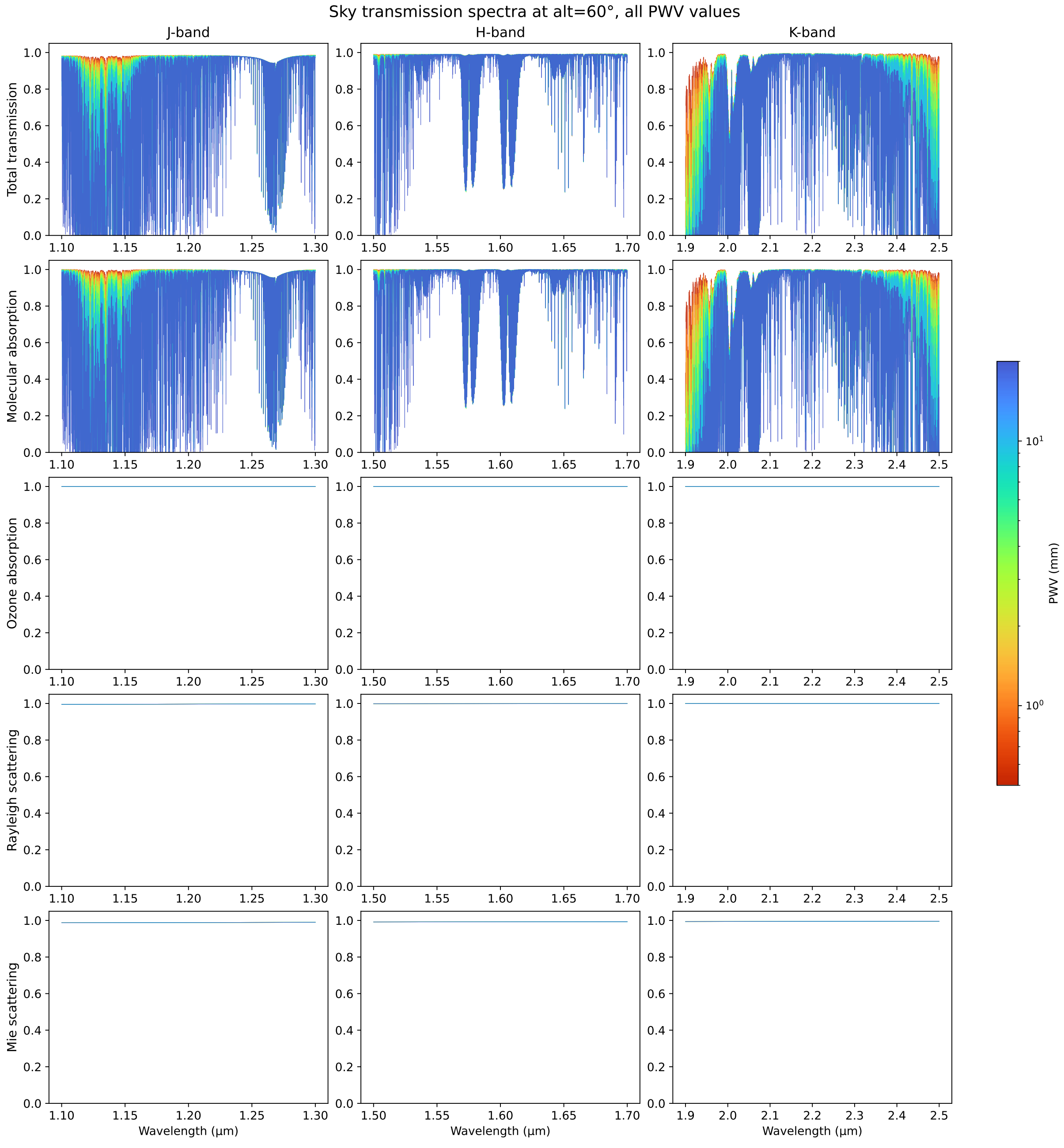

Figure A2: Transmission component spectra at altitude = 60° for all nine PWV values (coloured by PWV; drier = red, wetter = blue).

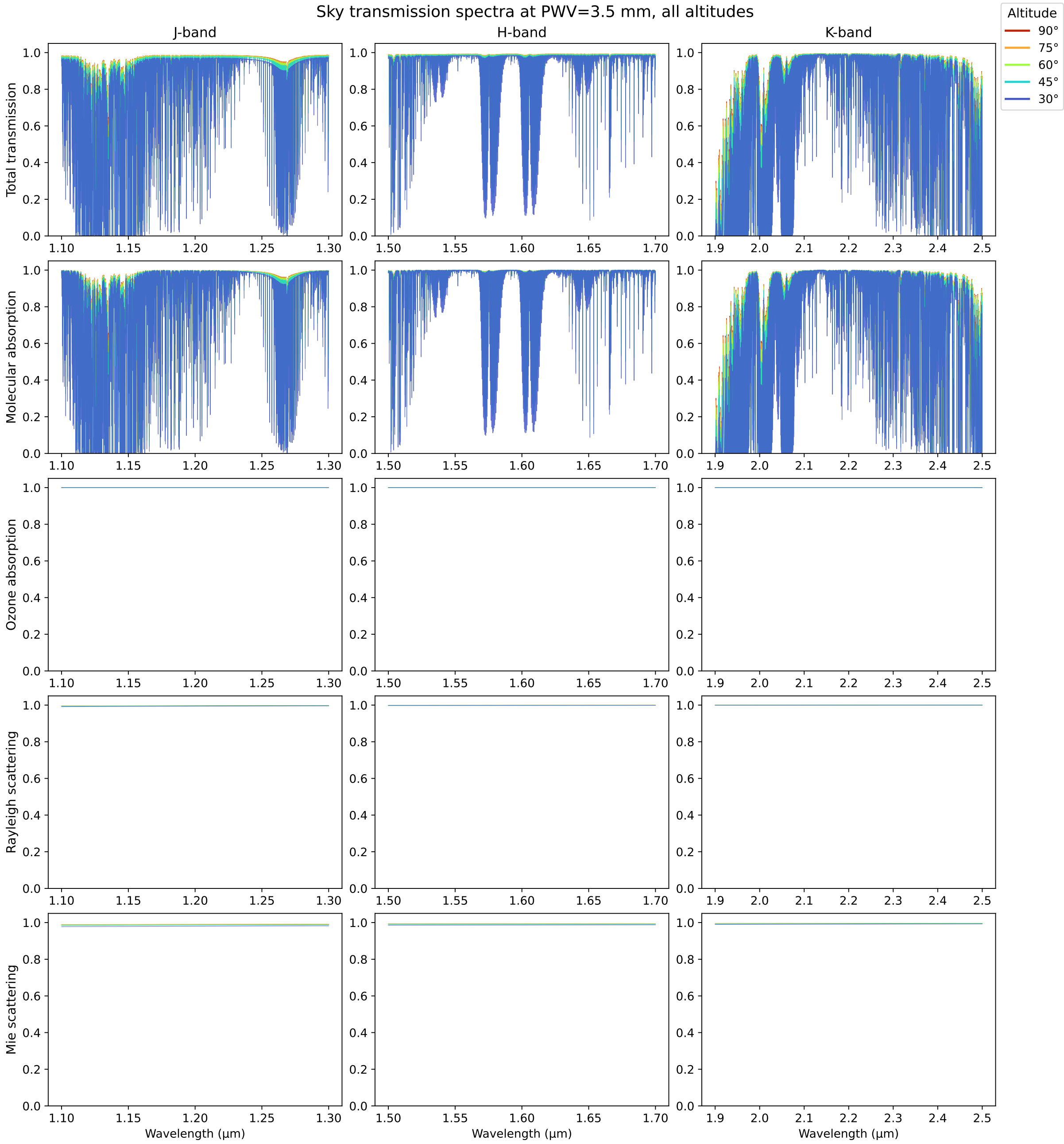

Figure A3: Transmission component spectra at PWV = 3.5 mm for all five altitudes (coloured by altitude; lower altitude = bluer).

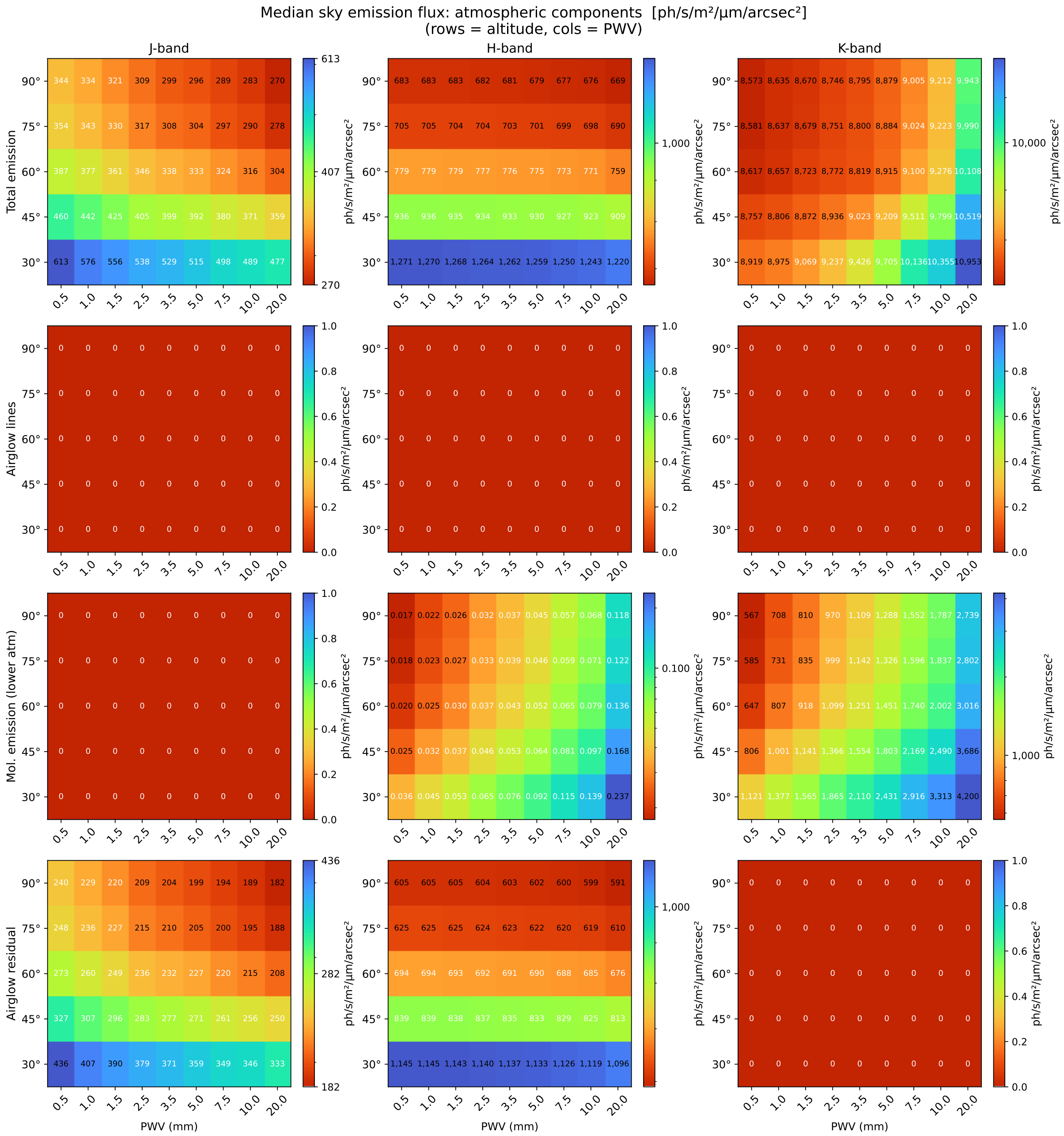

Figure A4: Median sky emission flux for atmospheric emission components (total, airglow lines, molecular emission, airglow residual) across the 5 × 9 altitude–PWV grid.

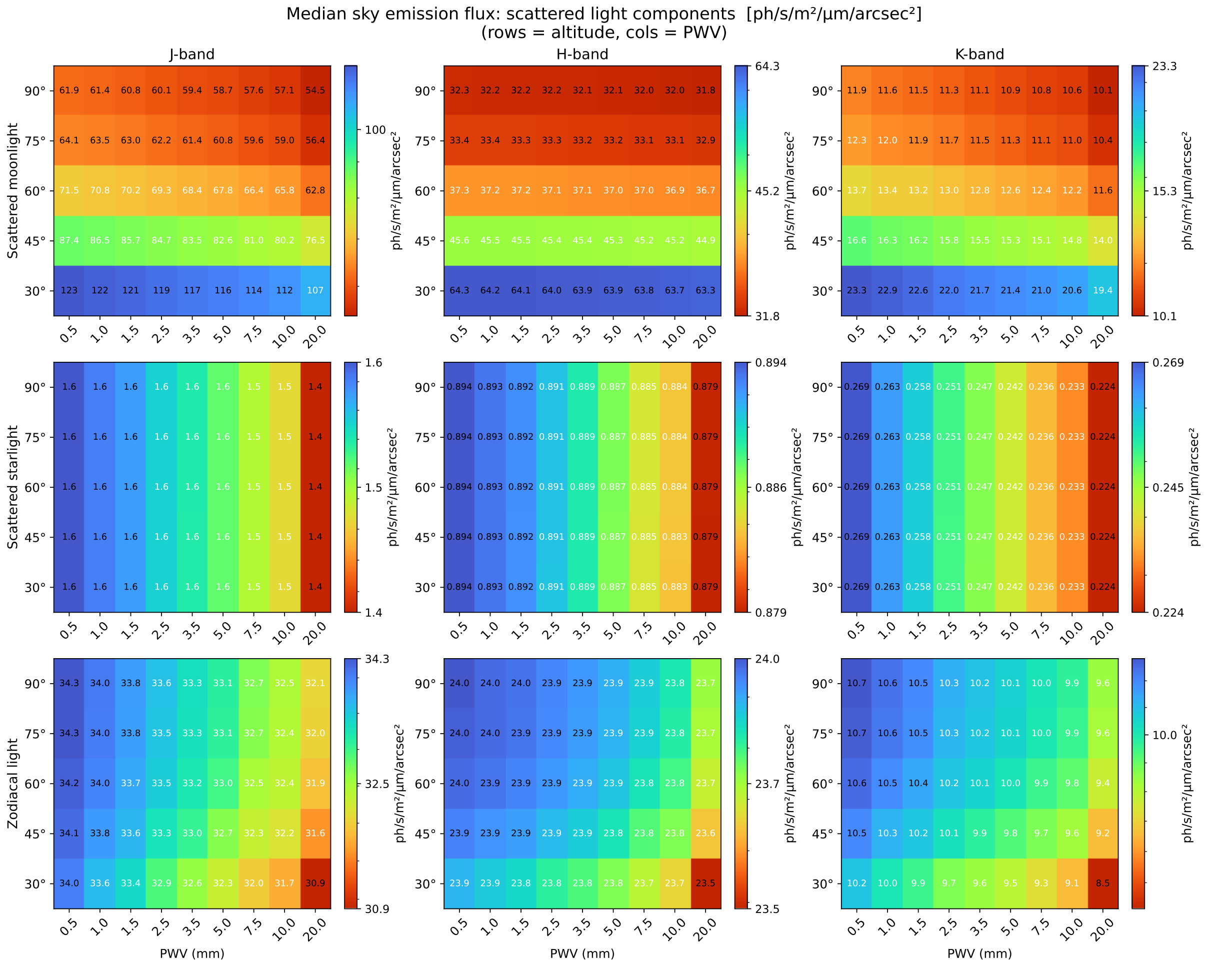

Figure A5: Median sky emission flux for scattered light components (moonlight, starlight, zodiacal light) across the 5×9 altitude–PWV grid.

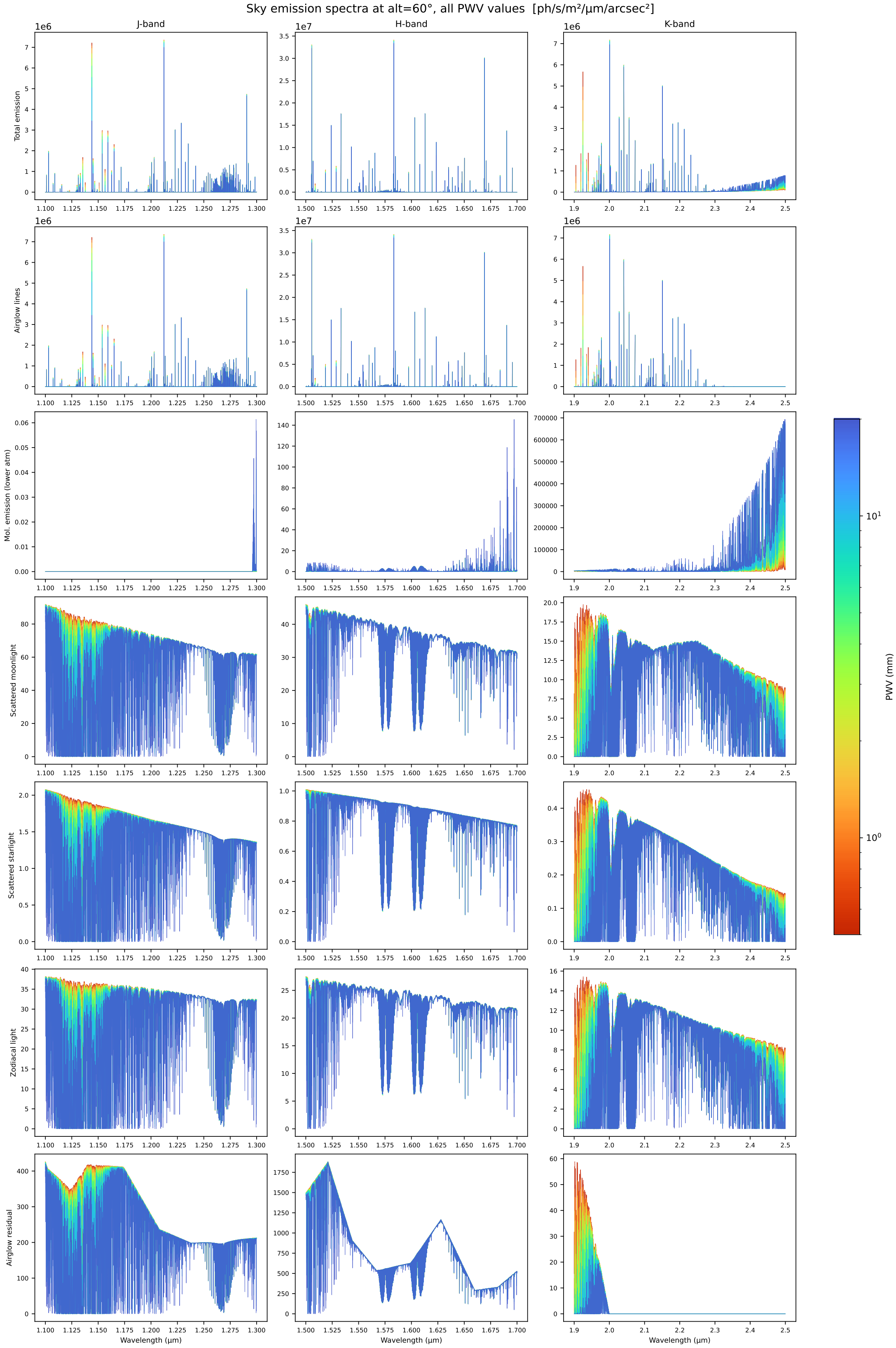

Figure A6: Sky emission component spectra at altitude = 60° for all nine PWV values.

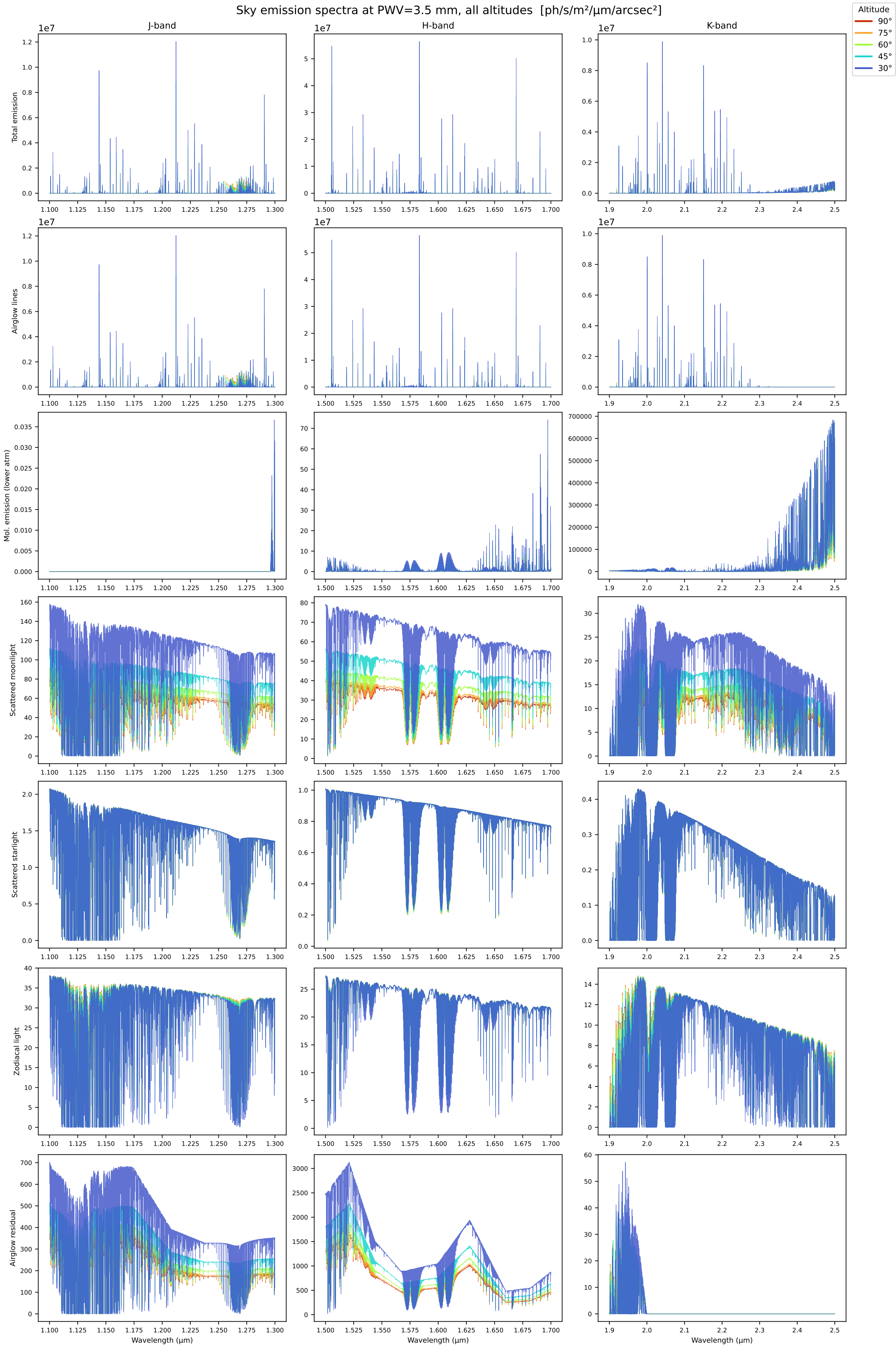

Figure A7: Sky emission component spectra at PWV = 3.5 mm for all five altitudes.

B Molecular Template Grid and Line Counting

B.1 Grid Overview

The ETC includes a precomputed grid of molecular templates covering five planet classes. Each template gives the absorption signature of a single molecular species as it would appear in a planetary transit spectrum. The templates are computed in a transmission geometry: starlight filtering through the planet's atmosphere at the day-night boundary acquires a set of narrow absorption features whose wavelength positions and relative depths are characteristic of each molecule. The user-selected planet class determines which set of templates is passed to the line-counting algorithm.

These templates are constructed for transmission spectroscopy, so their use for reflected-light observations is an approximation, as the physical geometry and resulting line depths differ between the two modes. The HRCCS detection estimate is nevertheless still meaningful because molecular line positions do not depend on the observing geometry. The `line_contrast` parameter is intended to account for this: it should be set by the user to reflect the expected planet-to-star flux ratio for the mode of interest. Users applying the reflected-light mode should treat the templates as a guide to where molecular lines fall in each band and set `line_contrast` accordingly. A future release will support user-supplied templates.

Two complementary radiative-transfer pipelines were used to populate the grid:

- **NASA Planetary Spectrum Generator (PSG)** (Villanueva et al., 2018)[2] configured with HITRAN line lists for the two temperate classes.
- **petitRADTRANS**[3] (Mollière et al., 2019) for the three hot classes where HITEMP / ExoMol opacity is essential to reproduce the high-temperature lines.

B.2 Planet Classes

Each class is defined by a representative equilibrium temperature and a set of molecular species. Table B1 summarises the grid. The per-class abundance set follows standard literature values for each class and can be provided upon request.

Table B1: Planet-class template grid. Templates are generated for each (class, molecule, band) triple in the J, H, and K windows of Table 7; the *Species* column lists which molecules are tabulated.

Class key	T_{eq}	**Species**
`temperate_terrestrial`	$\sim$250 K	CH_4, CO, CO_2, H_2O, O_2
`temperate_subneptune`	$\sim$270 K	CH_4, CO_2, H_2O, NH_3
`warm_subneptune`	$\sim$600 K	CH_4, CO, CO_2, H_2O, NH_3
`hot_jupiter`	$\sim$1500 K	CH_4, CO, CO_2, H_2O
`ultra_hot_jupiter`	$\sim$2500 K	CO, CO_2, H_2O, OH

B.3 Line Counting Algorithm

Lines are counted directly on the raw ppm template after linear interpolation onto the ETC wavelength grid:

[2] https://psg.gsfc.nasa.gov/
[3] https://petitradtrans.readthedocs.io/

1. Restrict the template to the selected band window $[\lambda_{\rm lo}, \lambda_{\rm hi}]$.
2. Compute the in-band dynamic range

$$\Delta T_{\rm ppm} \;=\; \max\!\big(T_{\rm ppm}\big) \;-\; \min\!\big(T_{\rm ppm}\big). \tag{B1}$$

3. Set the absolute prominence threshold

$$P_{\rm thr} \;=\; p_{\rm prom} \cdot \Delta T_{\rm ppm}, \tag{B2}$$

   where $p_{\rm prom}$ is the user input (default 0.05, i.e. 5% of the in-band range).
4. Invoke `scipy.signal.find_peaks` on the raw ppm template with `prominence` $= P_{\rm thr}$ and `distance` $= n_x$ pixels. `find_peaks` computes each peak's prominence locally as the height above the highest saddle out to the nearest taller peak; only peaks whose local prominence is $\geq P_{\rm thr}$ are retained.
5. Filter the surviving peaks by the local stellar SNR per resolution element: only peaks at wavelengths where $\mathrm{SNR}_{\rm res}(\lambda) \geq \mathrm{SNR}_{\rm thr}$ (default 100) are counted into $N_{\rm lines}$. The local SNR filter excludes lines falling in deep telluric absorption windows, where the stellar continuum is too low to support a usable cross-correlation contribution. The threshold is independent of the prominence cut; a peak passes only if both criteria are satisfied.

B.4 Caveats

The grid is intended to be representative, not exhaustive. Concrete limitations:

- Each class uses a single isothermal 1D profile. Thermal structure, photochemistry, and altitude-dependent abundances can modify line strengths by factors of a few.
- Clouds and hazes are not included. High-altitude aerosols suppress line depths, so cloudy targets will require more observing nights than the templates predict.
- Stellar contamination from starspots and faculae is not modelled and can be a leading systematic for active hosts.
- $N_{\rm lines}$ assumes each selected line contributes independently. Line blending and instrumental systematics modify this scaling. The parameter $p_{\rm detrend}$ (Equation (28)) provides a leading-order correction.

C Derivation of the HRCCS Detection SNR

This appendix derives the HRCCS detection SNR as a matched-filter problem. The aim is to estimate the amplitude of a known planetary template in a noisy residual spectrum. After detrending, the one-dimensional residual spectrum from a single exposure is written as

$$R_i = AT_i + n_i, \tag{C1}$$

where R_i is the residual flux at wavelength pixel i, T_i is a dimensionless template (Appendix B), A is the planet-signal amplitude, and n_i is zero-mean Gaussian noise with variance σ_i^2. The noise is assumed independent between pixels.

The planet amplitude is estimated by taking a weighted sum of the residual spectrum,

$$\hat{A} = \sum_i w_i R_i, \tag{C2}$$

where the weights w_i are chosen to give an unbiased estimate with minimum variance.

For this estimator to be unbiased, its expectation value must equal the true amplitude A. Assuming zero-mean noise and using Equation (C1) yields

$$\langle \hat{A} \rangle = A \sum_i w_i T_i. \tag{C3}$$

Therefore the weights must satisfy

$$\sum_i w_i T_i = 1. \tag{C4}$$

For independent noise, the variance of the estimator is

$$\mathrm{Var}(\hat{A}) = \sum_i w_i^2 \sigma_i^2. \tag{C5}$$

Among all choices of w_i that give an unbiased estimate of A, the weights that minimise the estimator variance are

$$w_i = \frac{T_i/\sigma_i^2}{\sum_k T_k^2/\sigma_k^2}. \tag{C6}$$

These weights give more importance to pixels where $|T_i|$ is large and the noise variance is small.

Substituting Equation (C6) into Equation (C2) gives the matched-filter estimator

$$\hat{A} = \frac{\sum_i T_i R_i/\sigma_i^2}{\sum_i T_i^2/\sigma_i^2}, \tag{C7}$$

with variance

$$\mathrm{Var}(\hat{A}) = \left(\sum_i \frac{T_i^2}{\sigma_i^2} \right)^{-1}. \tag{C8}$$

The corresponding standard uncertainty of the amplitude estimate is

$$\sigma_{\hat{A}} = \sqrt{\mathrm{Var}(\hat{A})} = \left(\sum_i \frac{T_i^2}{\sigma_i^2} \right)^{-1/2}. \tag{C9}$$

If the detrending procedure preserves only a fraction α of the planet signal, the recovered amplitude is αA. The corresponding CCF detection SNR is therefore

$$\mathrm{SNR}_{\mathrm{CCF}} = \frac{\alpha A}{\sigma_{\hat{A}}} = \alpha A \left(\sum_i \frac{T_i^2}{\sigma_i^2} \right)^{1/2}. \tag{C10}$$

Here α is a scalar approximation to the signal throughput of the detrending procedure. More generally, detrending can modify the shape of the recovered planet signal, in which case T_i should be interpreted as the effective template after injection and recovery through the same pipeline.

For correlated residuals, the diagonal noise variances are replaced by a full covariance matrix. With data vector $\mathbf{R}$, template vector $\mathbf{T}$, and covariance matrix $\mathbf{C}$, the matched-filter estimator becomes

$$\hat{A} = \left(\mathbf{T}^{\mathrm{T}}\mathbf{C}^{-1}\mathbf{T}\right)^{-1}\mathbf{T}^{\mathrm{T}}\mathbf{C}^{-1}\mathbf{R}, \tag{C11}$$

with variance

$$\mathrm{Var}(\hat{A}) = \left(\mathbf{T}^{\mathrm{T}}\mathbf{C}^{-1}\mathbf{T}\right)^{-1}. \tag{C12}$$

The corresponding detection SNR is

$$\mathrm{SNR}_{\mathrm{CCF}} = \alpha A \left(\mathbf{T}^{\mathrm{T}}\mathbf{C}^{-1}\mathbf{T}\right)^{1/2}. \tag{C13}$$

The independent-noise expression in Equation (C10) is recovered when $\mathbf{C}$ is diagonal with diagonal elements $C_{ii} = \sigma_i^2$.